

GeordieDaz AI

Technical Evidence Package

CONFIDENTIAL — CLIENT REVIEW

Prepared: 30 August 2026

1. OpenAI API — Credit Balance Exhausted

The following error was captured by sending a direct API request to OpenAI using the GeordieDaz project's API key on **30 August 2026 at 02:23 AM**.

Field	Value
Project	GeordieDaz AI Voice Assistant
API Key	sk-proj-d7-s.....Q5IA
Test Endpoint	gpt-4o-mini (cheapest available model)
Result	✗ BLOCKED — No Credits Remaining

Raw API Response from OpenAI

```
{ "error": { "message": "You have no credits remaining. Add credits to  
continue using the API at  
https://platform.openai.com/settings/organization/billing/.", "type":  
"insufficient_quota", "param": null, "code":  
"credit_balance_exhausted" } }
```

What This Means

- The API key is **valid** — OpenAI recognises and accepts it.
- The account has a **negative balance** which blocks all API calls.
- Every component of GeordieDaz (voice, memory, embeddings) requires active credits.

- The application code is fully deployed and operational — it is **only** blocked by the billing issue.

Application Error Flow

Step	Action	Result
1	User clicks "START SERVER" → Login	✓ Success
2	WebSocket connects to backend	✓ Success
3	LangGraph pre-turn pipeline runs	✓ Success
4	Backend connects to OpenAI Realtime API	✗ Rejected (insufficient_quota)
5	Error sent to frontend	"Failed to initialise voice session"

Resolution: Add credits at platform.openai.com/billing. The application will work immediately once the balance is positive.

2. Likely Credit Usage Breakdown

The -\$17.27 balance was most likely consumed by the following API services during development and testing:

Service	Cost	Usage
OpenAI Realtime API (Voice)	~\$0.06/min audio	Voice session testing, WebSocket reconnect loops during debugging
text-embedding-3-small	~\$0.02/1M tokens	Memory vector embeddings (each stored fact generates an embedding)
GPT-4o (Memory Extraction)	~\$2.50/1M input tokens	Extracting facts from conversation turns after each dialogue
GPT-4o-transcribe (Whisper)	~\$0.006/min	Speech-to-text transcription during voice sessions

Note: During debugging, repeated WebSocket reconnection loops (each attempt opens a new Realtime API session) likely consumed credits rapidly. This has been fixed — the application now handles errors gracefully without reconnect storms.

3. System Architecture Overview

Component	Technology	Hosting
Frontend (JARVIS HUD)	React + Vite + Canvas API	Vercel (free tier)
Backend API	FastAPI + Python	Render (web service)
Voice Engine	OpenAI Realtime API (WebSocket)	OpenAI Cloud
Voice Clone	ElevenLabs (Alan Robson clone)	ElevenLabs Cloud
Database	PostgreSQL + pgvector	Render (managed)
Cache / Working Memory	Redis	Render (managed)
Agent Pipeline	LangGraph (7 nodes)	Backend (in-process)
Authentication	JWT + bcrypt	Backend (in-process)

Repository Structure

```
geordiedaz/
├── backend/
│   ├── app/
│   │   ├── agent/nodes/      ← LangGraph pipeline (7 nodes)
│   │   ├── api/              ← REST endpoints (auth, memory, persona)
│   │   ├── models/           ← SQLAlchemy ORM (User, Memory, Session)
│   │   ├── services/         ← Business logic (memory, auth, TTS)
│   │   ├── ws/               ← WebSocket handler (OpenAI bridge)
│   │   ├── personas/         ← YAML persona configs
│   │   └── middleware/       ← JWT auth middleware
│   ├── alembic/              ← Database migrations
│   ├── tests/                ← 28 automated tests
│   └── requirements.txt
├── frontend/
│   ├── src/components/       ← React UI (JARVIS HUD)
│   ├── src/hooks/            ← WebSocket & Voice hooks
│   └── src/stores/           ← Zustand state management
├── docs/                     ← Character Bible
├── docker-compose.yml
└── README.md
```

